AABHA TAMHANKAR

Google Scholar | GitHub | LinkedIn | Website

EDUCATION

PhD in Robotics Engineering, Worcester Polytechnic Institute — *Expected* 2028
Thesis: Modeling, Control & Planning for Autonomous Neuroendovascular Procedures — GPA: 4.00

Master's in Robotics Engineering, Worcester Polytechnic Institute — May 2024
Thesis: Image Guidance and Safe Insertion Region Localization for Lumbar Puncture — GPA: 3.88

RESEARCH EXPERIENCE

Autonomous Neuroendovascular Procedures — Aug 2024 – Present
PhD Student — FuTURE Lab — Project Link

- Architecting full-stack **patient-specific vascular digital twins** for autonomous neurovascular surgical interventions.
- Collaborating with neurosurgeons at UMass Medical and Boston Children's Hospital on optimal navigation methods.
- Developing GPU-accelerated **physics-based simulation models** to capture contact mechanics for flexible tools.
- Achieved a **100% success rate** in closed-loop guidewire-catheter navigation across 600+ simulated aortic anatomies.
- Integrated a **custom robotic tool actuator** with **live C-arm** imaging and path planning via **ROS2 pipeline.**
- Resolving **monoplane fluoroscopy** depth ambiguity to accurately localize 3D tool poses for navigation feedback.
- Developing SOFA-driven **multi-objective optimization** to select optimal tool mechanics for patient-specific anatomies.

Image Guidance for Lumbar Puncture — Aug 2023 – May 2024
Master's Thesis Research — Thesis Link

- Built an **MRI–ultrasound multimodal registration** framework to enhance 3D intraoperative visualization.
- Implemented **intensity-based segmentation and 3D reconstruction** pipelines for live 2D ultrasound streams.
- Engineered real-time **3D spatial tracking** for the US probe, enabling intraoperative safe insertion region mapping.
- Improved registration accuracy by **13.2% IoU** across 44 dynamic test cases using custom image-based local optimization.

PUBLICATIONS

- **A. Tamhankar**, S. Verma, J. Patil, G. Pittiglio, *"Contact-aware Path Planning and Online Re-planning for Neuroendovascular Guidewire Navigation with Single Plane Fluoroscopic Feedback,"* in **IEEE Transactions on Medical Robotics and Bionics (T-MRB)**, *Under Review.*
- **A. Tamhankar**, J. Patil, G. Pittiglio, *"An Anatomy-specific Guidewire Shaping Robot for Improved Vascular Navigation,"* in *Proc. IEEE International Symposium on Medical Robotics (ISMR)*, 2026.
- **A. Tamhankar**, R. Alterovitz, A. Puri, G. Pittiglio, *"Contact-aware Path Planning for Autonomous Neuroendovascular Navigation,"* in **IEEE Robotics & Automation Letters (RA-L)**, 2025.
- **A. Tamhankar**, G. Pittiglio, *"Towards Autonomous Navigation of Neuroendovascular Tools for Timely Stroke Treatment via Contact-Aware Path Planning,"* in *Proc. IEEE International Symposium on Medical Robotics (ISMR)*, 2025. **[2nd Place Best Student Paper Award].**

WORK EXPERIENCE

Engineering Development Group Intern, MathWorks *Natick, MA* — May 2023 - Aug 2023

- Enhanced **3D mapping and obstacle localization** by improving TSDF representation via voxel-level curve fitting.
- Achieved **15% higher accuracy** and **12% runtime cut** over baseline TSDF in MATLAB driving simulations.

ACHIEVEMENTS & SERVICE

- **1st Place, Stäubli Future Medical Roboticists Award** for research on contact-aware path planning at the *"Training the Future Medical Roboticists"* workshop, IROS 2026.
- **2nd Place, Best Student Paper Award** at IEEE Intl. Symposium on Medical Robotics (ISMR), 2025.
- Winner, People's Choice Award at Graduate Research Innovation Exchange (GRIE) WPI, 2024.
- Invited Speaker at WPI Robotics Colloquium: *"Towards Autonomous Neuroendovascular Interventions"*.
- Graduate Lecturer on **Optical Flow** and **Motion Segmentation** for RBE 549 for a cohort of ≈60 students.
- Serving as a research mentor providing technical guidance to 10+ graduate students at the FuTURE Lab.